SIMATS ENGINEERING

ASSESSMENT TOOL 1: Analytical Problem Solving

COURSE NAME: Cloud Computing and Big Data Analytics **COURSE CODE:** CSA15

COURSE OUTCOME COVERED

CO2: *Analyze virtualization architectures to identify performance and security issues in cloud computing environments.* (BL3)

Assessment Tool Weightage: 40%

Dave's Taxonomy: Manipulation (Level 2) **Total Marks: 50**
Question Count: 5

Code of Conduct

I Sanjay S/192521058 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: Sanjay S

Assessment Tasks

Analytical Problem Solving

1. Compare Type-1 and Type-2 hypervisors for a cloud datacenter hosting mission-critical applications. Analyze performance, security, and manageability, then justify the better choice.
2. A cloud provider must isolate workloads belonging to finance, healthcare, and student portals on the same hardware. Analyze how virtualization architecture supports isolation and identify two major attack surfaces.
3. Study the following VM host utilization snapshot and recommend corrective actions: CPU 92%, memory 88%, storage I/O latency 35 ms, average VM boot time 170 s. Explain the likely bottlenecks.
4. Analyze how Software Defined Datacenter architecture improves agility compared with a traditional datacenter. Support your answer with compute, storage, and network virtualization considerations.
5. A multi-cloud federation project fails due to identity mismatches between providers. Analyze the issue and propose a secure identity and privacy design using federation concepts.

Analytical Problem Solving Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 – Proficient	Level 2 - Developing	Level 1 - Beginning
Problem Interpretation	20%	Interprets all technical requirements accurately	Interprets most requirements correctly	Partial interpretation	Misinterprets the problem
Analytical Reasoning	25%	Strong reasoning with clear cause-effect analysis	Good reasoning with minor gaps	Basic analysis only	Weak or no analysis
Method Selection	20%	Chooses highly appropriate virtualization and security concepts	Chooses mostly suitable concepts	Partially suitable concepts	Inappropriate approach
Solution Quality	20%	Provides feasible and technically sound solution	Reasonable solution with minor issues	Basic solution with limitations	Unclear or invalid solution
Presentation of Analysis	15%	Well-structured and technically precise	Generally clear	Somewhat unclear	Difficult to follow

Total Marks Awarded:

Compare Type-1 and Type-2 hypervisors for a cloud datacenter hosting mission-critical applications. Analyze performance, security, and manageability, then justify the better choice

1. Problem Interpretation

A cloud datacenter hosting mission-critical applications must provide high performance, strong security, high availability, and easy management. Hypervisors enable multiple virtual machines (VMs) to run on a single physical server.

- Type-1 Hypervisor (Bare-Metal): Runs directly on the physical hardware without requiring a general-purpose host operating system.
- Type-2 Hypervisor (Hosted): Runs as an application on top of a host operating system.

The objective is to compare both hypervisors based on performance, security, and manageability and select the most suitable option for mission-critical cloud applications.

2. Analytical Reasoning

Factor	Type-1 Hypervisor	Type-2 Hypervisor
Performance	High; direct access to hardware	Relatively lower due to host OS overhead
Security	Smaller attack surface; fewer software layers	Additional host OS increases potential vulnerabilities
Manageability	Centralized management for large-scale deployments	Easier for individual machines and testing
Reliability	Better suited to continuous workloads	Depends on the stability of the host OS
Scalability	Supports large cloud infrastructures	Less suitable for large production datacenters
Examples	VMware ESXi, Microsoft Hyper-V, Xen	Oracle VirtualBox, VMware Workstation

Analysis:

- Performance: Type-1 hypervisors reduce virtualization overhead and provide efficient CPU, memory, and I/O resource management.
- Security: Type-1 hypervisors have fewer software layers between hardware and virtual machines, potentially reducing the attack surface. However, proper patching, configuration, and access control are still essential.
- Manageability: Type-1 hypervisors support centralized administration, resource allocation, monitoring, and integration with high-availability systems.
- Reliability: Type-1 deployments are generally better suited to workloads that require continuous operation and predictable performance.

3. Method Selection

The appropriate method is a comparative decision analysis using three primary criteria: performance, security, and manageability.

For mission-critical applications, the selected hypervisor should prioritize low overhead, isolation, centralized management, and high availability.

A weighted evaluation can be used:

Criterion	Weight	Type-1 Score (out of 5)	Type-2 Score (out of 5)
Performance	40%	5	3
Security	35%	4	3
Manageability	25%	5	3
Weighted total	100%	4.65/5	3.00/5

Note: Scores are illustrative estimates for this scenario, not benchmark measurements.

4. Solution Quality

Recommended solution: Type-1 hypervisor.

Type-1 hypervisors are the better choice for a mission-critical cloud datacenter because they provide:

1. High performance with lower virtualization overhead.
2. Stronger isolation through a reduced software stack.
3. Centralized management of virtual machines and physical resources.
4. Better integration with clustering, live migration, and high-availability mechanisms.
5. Greater suitability for scaling enterprise and cloud workloads.

For example, a healthcare cloud system that stores patient records and supports appointment services needs continuous availability, reliable performance, and strong access control. A Type-1 hypervisor combined with redundant servers, backups, monitoring, and disaster recovery can support these requirements.
Type-2 hypervisors remain useful for development, learning, testing, and desktop virtualization, but are generally less appropriate for production cloud datacenters.

5. Presentation of Analysis
Final Decision
Type-1 Hypervisor
Recommended for mission-critical cloud datacenters
Best overall fit for performance, security, scalability, and enterprise management.
Conclusion: Type-1 hypervisors are preferable for mission-critical cloud environments because they run directly on physical hardware, reduce host operating system overhead, and support enterprise-grade management and availability features. Therefore, they provide a stronger foundation for reliable cloud infrastructure than Type-2 hypervisors.

2.A cloud provider must isolate workloads belonging to finance, healthcare, and student portals on the same hardware. Analyze how virtualization architecture supports isolation and identify two major attack surfaces.

1. Virtualization-Based Workload Isolation
Problem Interpretation
A cloud provider hosts finance, healthcare, and student portal applications on the same physical hardware. The challenge is to ensure that each workload remains isolated so that unauthorized access, data leakage, or interference between applications does not occur.
Analytical Reasoning
Virtualization divides a physical server into multiple virtual machines (VMs). Each VM can run its own operating system, applications, and security policies.
Shared Physical Server
Hypervisor — Isolation and Resource Control
Finance VM
Financial data
Healthcare VM
Patient records
Student VM
Academic data
The hypervisor controls CPU, memory, storage, and device access. Separate virtual networks, access controls, and storage permissions provide additional protection.
Two major attack surfaces:

1. Hypervisor attack surface: Vulnerabilities in the hypervisor or virtual device emulation may allow a compromised VM to escape its isolation and affect the host or other VMs.
2. Virtual network attack surface: Misconfigured virtual switches, exposed management interfaces, or weak network access controls may permit unauthorized communication or data interception between workloads.

Method Selection
Use Type-1 hypervisor isolation, separate virtual networks, least-privilege access, encryption, and regular security patching.
Solution Quality
This architecture improves tenant isolation and resource control. However, VM separation alone is insufficient; network policies, monitoring, and secure configuration are also necessary.
Presentation of Analysis
Conclusion: Virtualization enables multiple organizations to share physical hardware while maintaining logical separation. Hypervisor vulnerabilities and virtual network misconfigurations are two critical risks that must be

mitigated.

3.Study the following VM host utilization snapshot and recommend corrective actions: CPU 92%, memory 88%, storage I/O latency 35 ms, average VM boot time 170 s. Explain the likely bottlenecksProblem Interpretation

The cloud host shows high resource utilization and slow VM startup. The objective is to identify the likely bottlenecks and recommend corrective actions.

Analytical Reasoning

Metric	Observed value	Interpretation
CPU utilization	92%	Very high; limited processing headroom
Memory utilization	88%	High; possible memory contention
Storage I/O latency	35 ms	Elevated for many latency-sensitive workloads
Average VM boot time	170 seconds	Slow; may reflect resource contention or startup overhead

Likely bottlenecks:

- CPU: Competing VMs may be consuming most available processing capacity, causing CPU scheduling delays.
- Memory: Memory pressure can cause ballooning, swapping, or host paging, which may degrade performance.
- Storage: High I/O latency can delay virtual disk reads and writes, including operating system startup operations.
- VM boot time: Startup may be delayed by storage bottlenecks, CPU contention, memory pressure, or simultaneous boot requests.

Method Selection

Use performance monitoring and bottleneck analysis. Examine CPU ready time, memory swapping, storage queue depth, IOPS, and boot logs before changing the configuration.

Solution Quality — Corrective Actions

Priority	Recommended action	Expected benefit
1	Identify and stop unnecessary or runaway workloads	Reduces CPU contention
2	Add RAM or rebalance VM memory allocations	Reduces memory pressure and swapping
3	Investigate storage queues; use faster SSD/NVMe storage if justified	Reduces I/O latency
4	Spread workloads across additional hosts	Improves capacity and availability
5	Stagger VM startup and use templates or optimized images	Reduces startup bursts and boot time

Presentation of Analysis

Conclusion: CPU utilization of 92% and memory utilization of 88% indicate limited resource headroom, while 35 ms storage latency suggests a possible I/O bottleneck. The first step is to correlate these metrics with VM boot activity, then relieve the confirmed bottleneck. Adding resources without diagnosing the cause may not solve the problem.

4. Analyze how Software Defined Datacenter architecture improves agility compared with a traditional datacenter. Support your answer with compute, storage, and network virtualization considerations

A traditional datacenter relies heavily on manually configured physical servers, storage devices, and network equipment. The goal is to analyze how an SDDC improves agility through software-controlled infrastructure.

Analytical Reasoning

An SDDC virtualizes compute, storage, and networking and manages them through software, policies, and automation.

Software-Defined Control and Automation

Policies • APIs • Orchestration • Monitoring

Compute

VMs and resource scheduling

Storage

Virtual pools and policies

Network

Virtual networks and security

Compute virtualization: VMs can be provisioned, resized, and migrated through software instead of manually configuring each server.

Storage virtualization: Physical disks are combined into logical storage pools. Capacity and performance policies can be applied centrally.

Network virtualization: Virtual switches, software-defined networks, and automated security rules enable rapid network provisioning and workload isolation.

Compared with traditional infrastructure, these capabilities reduce manual configuration, improve resource utilization, and accelerate service deployment.

Method Selection

Use a comparative architectural analysis across provisioning speed, resource allocation, scalability, and automation.

Factor	**Traditional datacenter**	**SDDC**
Provisioning	Often manual	Automated through software
Compute	Hardware-focused	Virtualized and pooled
Storage	Separate device management	Software-managed storage pools
Networking	Manual device configuration	Programmable virtual networks
Scaling	May require lengthy changes	Faster policy-driven scaling

Solution Quality

An SDDC improves agility, consistency, and operational efficiency. However, it requires skilled administrators, robust automation, compatible hardware, and strong control-plane security. Automation errors can also affect many resources at once.

Presentation of Analysis

Conclusion: SDDC architecture is more agile than a traditional datacenter because compute, storage, and networking are managed as programmable resources. It enables faster deployment, dynamic scaling, and consistent policy enforcement.

5. A multi-cloud federation project fails due to identity mismatches between providers. Analyze the issue and propose a secure identity and privacy design using federation concepts

Problem Interpretation

A multi-cloud project fails because different providers use inconsistent identity formats, user identifiers, roles, and authentication methods. The objective is to establish secure cross-cloud authentication while preserving privacy and access control.

Analytical Reasoning

Identity federation allows users authenticated by a trusted Identity Provider (IdP) to access authorized services at other providers without maintaining separate passwords for each cloud.

Enterprise Identity Provider

Central authentication and MFA

Federation Layer

SAML 2.0 / OpenID Connect • Claim mapping • Trust validation

Cloud A

Mapped roles

Cloud B

Mapped roles

Cloud C

Mapped roles

Why identity mismatches occur:

- One provider identifies users by email, while another uses a unique subject identifier.
- Role names and permission models differ between providers.
- Token formats, issuer values, audiences, or trust configurations do not match.
- User attributes and group memberships may be mapped incorrectly.

Secure identity and privacy design:

1. Establish a trusted IdP: Use a central enterprise identity service with multi-factor authentication (MFA).
2. Adopt federation standards: Use OpenID Connect (OIDC) or SAML 2.0, according to provider support.
3. Normalize identity claims: Map stable user identifiers, groups, and attributes into each cloud's roles.
4. Enforce least privilege: Apply role-based or attribute-based access control and grant only required permissions.
5. Protect tokens: Validate signatures, issuer, audience, expiry, and other required claims; use short-lived tokens and secure key rotation.
6. Preserve privacy: Share only necessary identity attributes, encrypt communications, and maintain appropriate audit logs and retention policies.
7. Monitor access: Revoke access when roles change or accounts are disabled, and review cross-cloud permissions regularly.

Method Selection

Use a standards-based identity federation architecture with centralized authentication, provider-specific role mapping, and least-privilege authorization.

Solution Quality

This approach reduces duplicate accounts and password management while making cross-cloud access more consistent. Its success depends on correct trust configuration, accurate claim mapping, token validation, and reliable account revocation.

Presentation of Analysis

Conclusion: Identity federation provides a secure way to authenticate users across multiple cloud providers. A trusted IdP, OIDC or SAML, normalized identity claims, least-privilege authorization, and data minimization can resolve identity mismatches while protecting user privacy.